

The VGG19 and DenseNet121 Model Comparison for the Chest X-Ray Based Tuberculosis Detection on Local Dataset

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Abstract—Globally, tuberculosis (TB) remains a significant infectious disease conducting massive morbidity and demanding more accessible diagnostic tools. Chest X-ray (CXR) interpretation, a standard screening modality that is highly dependent on interpreter expertise and prone to inter-reader variability, needs computer-aided diagnosis (CAD) systems to address these challenges. This study compares two convolutional neural network (CNN) models, VGG19 and DenseNet121, as a potential method for CAD TB detection from CXR images on a local dataset of 4,784 images. Both architectures were initialized with ImageNet-pretrained weights to ensure a consistent transfer learning approach, and DenseNet121 was further trained with class-weighted loss, a weighted random sampler, data augmentation, and a learning rate scheduler to address class imbalance and improve generalization. We set up each model to vary the batch size, with 30 epochs and Adam as the optimizer. Based on the experimental results, DenseNet121 outperformed VGG19 across all batch sizes, achieving ROC-AUC values between 0.790 and 0.823 compared to 0.557–0.735 for VGG19. Analysis of training and validation curves revealed that overfitting in DenseNet121 became more pronounced as batch size increased, with the largest train-validation gap observed at batch size 16 and the smallest at batch size 4. Furthermore, the performance metrics, such as accuracy, recall, precision, and F1-score, for DenseNet121 and VGG19 testing show the superiority of DenseNet over VGG. The performance comparison between the VGG19 and DenseNet121 CNN architectures for TB detection from CXR images is highly sensitive to the choice of batch size during training. These findings suggest that appropriate training strategies, including batch size selection and regularization techniques, are critical for improving the generalization of CNN models.

Index Terms—cnn, cxr, densenet, tuberculosis, vgg

I. INTRODUCTION

Indonesia needs a system that supports early detection and well-distributed devices to accelerate the reduction of tuberculosis (TB) incidence nationwide. Similar to the global situation, the burden of TB in Indonesia remains high and continues to pose a major public health challenge [1]–[3]. TB, caused by *Mycobacterium tuberculosis*, primarily affects the pulmonary system, especially the lungs, although it can also involve other organs. According to the World Health Organization (WHO) report in 2022, the global incidence of TB was approximately 10.6 million cases, with an estimated

mortality of approximately 1.3 million deaths [1], [2]. They also reported TB as one of the leading causes of death from infectious diseases, ranking below COVID-19, even surpassing HIV [1], [2]. While Indonesia is among the countries with the highest TB burden worldwide, with a substantial number of undetected cases [3]. Therefore, rapid and accurate early detection is essential to interrupt transmission chains and ensure timely and appropriate treatment.

CXR imaging, as one of the most widely used screening modalities, offers several advantages in detecting pulmonary abnormalities, especially those associated with TB. It includes its non-invasive nature, relatively rapid acquisition and analysis, cost-effectiveness, and wide accessibility. CXR can be used to identify pulmonary features, including upper lobe infiltrates, cavitation, pleural effusions, and lymphadenopathy [4], which are valuable for rapid and accurate TB screening. However, the accuracy of CXR interpretation depends heavily on the expertise and experience of the radiologist or clinician performing the assessment. Several studies have reported that inter-observer variability in CXR-based TB diagnosis remains substantial, particularly in regions with limited access to experienced radiological professionals [5], [6]. Therefore, this limitation has driven the development of computer-assisted diagnostic systems to improve the consistency and accuracy of clinical decision-making.

Over the past decade, artificial intelligence (AI) has advanced significantly, particularly in deep learning. Thereby, it brought substantial opportunities for affordable and accurate medical image analysis. Convolutional Neural Networks (CNNs) have emerged as one of the most prominent methods for processing image-based data [7]. Numerous prior studies have demonstrated that CNNs can achieve superior performance across a wide range of computer vision tasks, including the classification and detection of abnormalities in radiological images [8], [9]. This is largely due to their ability to learn hierarchical feature representations directly from raw pixel data [8], [9]. Their ability to automatically extract relevant features without manual feature engineering makes them particularly suitable for handling complex, highly heterogeneous CXR

images. Several well-established CNN architectures, such as VGG and DenseNet, have been widely used in previous studies to support TB screening via CXR image analysis [10]–[13]. In particular, VGG19, with 19 weight layers, is known for its strong feature extraction and consistent performance across benchmark datasets, making it a popular choice for medical image classification [14]. Meanwhile, DenseNet121, a CNN model introduced by [15], has also been shown to be an effective model for medical image classification, especially in scenarios with limited training data. Therefore, a comparative evaluation of these two architectures on a local dataset constitutes the primary novelty of this study.

The aims of this study are to evaluate the effectiveness of the CNN model and to compare the VGG19 and DenseNet121 architectures for TB classification on CXR images from a local dataset collected at a pulmonary hospital in Indonesia. During the study, we designed experiments to investigate the effect of batch-size variation during model development on classification performance, which was assessed through validation and testing procedures for each architecture. In addition to reporting performance metrics, this study emphasizes a critical analysis of the clinical implications of each experimental outcome, aiming to provide insights for future model optimization. The main contributions of this study are as follows: 1) the implementation of VGG19 and DenseNet121 models initialized with ImageNet-pretrained weights for binary TB classification, ensuring a consistent transfer learning strategy across both architectures, 2) the analysis of the impact of batch size tuning on the performance and generalization behavior (including overfitting tendencies) of both architectures, and 3) a comparative evaluation of VGG19 and DenseNet121 in classifying TB using a local dataset, supported by an analysis of training-validation dynamics across batch sizes.

II. METHODS

This section presents the method and materials used for this study. The experimental design used three configurations (Test 1, Test 2, and Test 3) for each architecture applied to the local CXR dataset.

A. System Design

Fig. 1 illustrates the research workflow developed in this study. It consists of several stages, including data preprocessing, data splitting, model development, and model evaluation. Firstly, 4,784 CXR images were collected from a pulmonary hospital in Indonesia. They have been annotated by radiologists as TB or non-TB, comprising 3,142 and 1,642 images, respectively. This dataset composition indicates a class imbalance between the two categories, reflecting the real clinical conditions at the data collection site. Therefore, multiple performance metrics were employed during the evaluation stage to account for the impact of this imbalanced dataset on model performance.

The dataset is then preprocessed to ensure compatibility with the CNN model. In this stage, each image is resized to

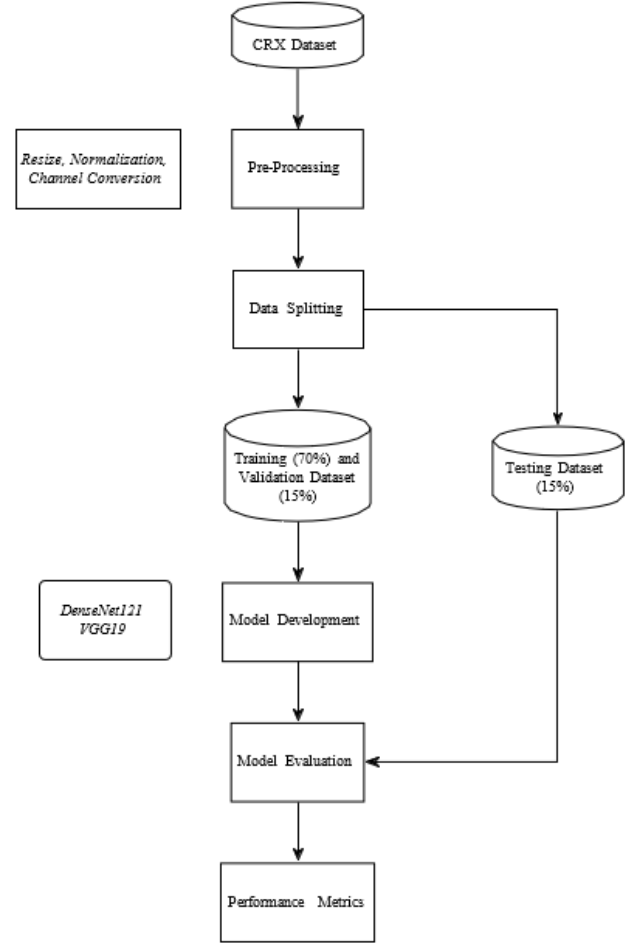


Fig. 1. Proposed research workflow for CXR-based tuberculosis classification.

224×224 pixels, min-max normalized to the range $[0, 1]$ to improve training stability, and mean-subtracted using ImageNet statistics. For the next stage, the dataset is divided into training, validation, and test sets using a stratified approach to maintain class proportions across the three experimental configurations. We split the data into a 70:15:15 ratio for training, validation, and testing, respectively. The training dataset is used to train both architectures during model development, while the validation dataset is used to monitor the learning process. To analyze the impact of batch size on model performance, three experiments are conducted with different batch sizes (4, 8, and 16) while maintaining the same learning rate and optimizer for each architecture. Finally, the trained model is evaluated using the testing dataset. Performance metrics such as accuracy, precision, recall, F1-score, ROC Curve and confusion matrix are used to assess the classification performance of the proposed system.

B. Model Architecture

VGG19 is a deep CNN architecture consisting of 16 convolutional layers and 3 fully connected layers, totaling 19 trainable weight layers. The network employs small 3×3

convolutional kernels throughout and uses max-pooling for spatial downsampling [14]. In this study, the VGG19 model was initialized with ImageNet-pretrained weights to leverage transfer learning, enabling the network to leverage rich feature representations learned from large-scale natural image data. The original classification head was replaced with a custom binary classifier comprising a Global Average Pooling layer, a Dense layer with ReLU activation, a Dropout layer for regularization, and a final sigmoid output neuron for binary TB classification. Fine-tuning was applied to the upper convolutional blocks during training to adapt the model to the CXR domain. The VGG architecture is illustrated in Fig. 2 [16].

Meanwhile, we chose DenseNet121 for its efficient reuse of features through dense connections, which are particularly useful for medical datasets that aren't massive in scale [15]. The DenseNet architecture is illustrated in Fig. 3 [16]. The network consists of an initial convolutional layer, four dense blocks separated by transition layers, a global average pooling layer, and a classification layer. In line with the VGG19 configuration, DenseNet121 was also initialized with ImageNet-pretrained weights to ensure a methodologically consistent comparison between the two architectures, eliminating initialization asymmetry as a confounding factor. The original classifier was replaced with a sequential head consisting of a fully connected layer (512 units) with ReLU activation, a dropout layer (rate = 0.5) for regularization, and a final linear layer with two output units for binary classification.

To address the class imbalance present in the dataset, two complementary strategies were employed: (1) class-weighted cross-entropy loss, where class weights were computed using median frequency balancing, and (2) a weighted random sampler during training, which increased the sampling probability of the minority (non-TB) class. The training process uses the Adam optimizer with weight decay (1×10^{-4}) for regularization, and a learning rate of 3×10^{-4} . A ReduceLROnPlateau scheduler was applied, reducing the learning rate by a factor of 0.5 when the validation loss failed to improve for 3 consecutive epochs, enabling adaptive convergence and serving as an early-warning mechanism against prolonged stagnation.

1) *Data Augmentation*: To improve model generalization and reduce overfitting risk, data augmentation was applied exclusively to the training set, while the validation and test sets retained only resizing and normalization to preserve evaluation integrity. The augmentation pipeline consisted of: (1) random horizontal flipping with a probability of 0.5, to account for left-right anatomical variation in chest radiographs; (2) random rotation within $\pm 10^\circ$, to simulate minor positioning variability during image acquisition; and (3) random resized cropping with a scale range of 0.9–1.0, to introduce slight scale and framing variation. All images were subsequently resized to 224×224 pixels and normalized using ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]), consistent with the requirements of ImageNet-pretrained backbones.

These augmentation strategies were selected to introduce

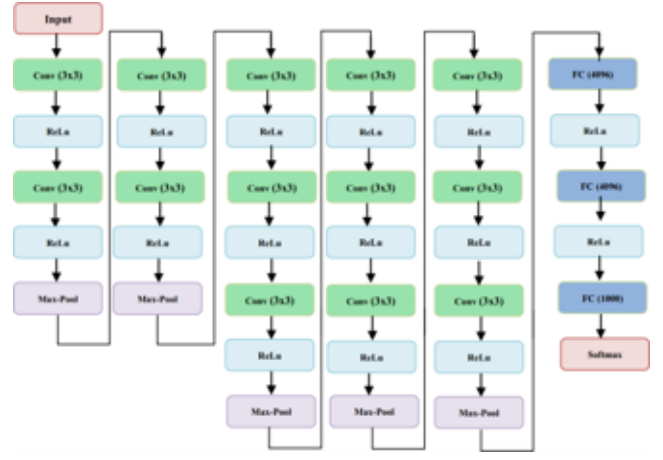


Fig. 2. Architecture diagram of VGG.

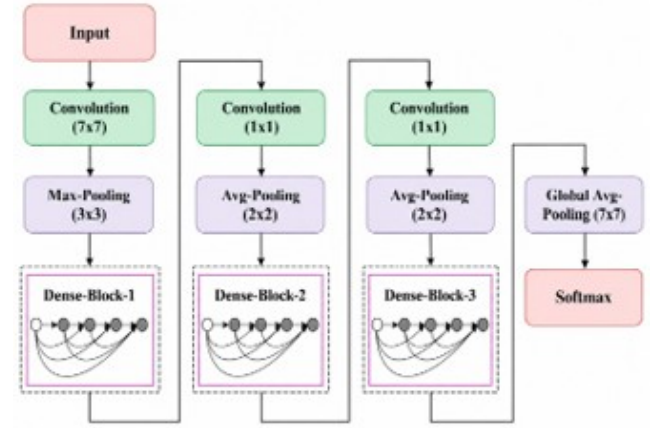


Fig. 3. Architecture diagram of DenseNet.

realistic variability commonly found in clinical CXR acquisition (patient positioning, slight rotation, framing) without distorting diagnostically relevant features such as lung field boundaries or opacity patterns. By exposing the model to a broader range of plausible input variations during training, augmentation is intended to discourage the network from memorizing dataset-specific artifacts and to encourage learning of more generalizable representations, thereby contributing to the mitigation of overfitting discussed in Section III-D3.

C. Experimental Setup

Three experimental configurations were implemented to systematically investigate the impact of mini-batch sizes on the model's sensitivity-specificity trade-off and training stability. The experiments were defined solely by their batch sizes: Test 1 used a batch size of 4, Test 2 used a batch size of 8, and Test 3 used a batch size of 16, illustrated by Table I.

All experiments were trained for 30 epochs using the cross-entropy loss function with the Adam optimizer. Training and validation loss/accuracy curves were monitored to assess learning dynamics. Model performance was evaluated using

TABLE I
EXPERIMENT CONFIGURATION FOR EACH ARCHITECTURE ON MODEL DEVELOPMENT

Experiment	Batch Size	Epoch	Learning Rate	Optimizer
Exp-1	4	30	5×10^{-5}	Adam
Exp-2	8	30	5×10^{-5}	Adam
Exp-3	16	30	5×10^{-5}	Adam

TABLE II
EVALUATION REPORT FOR EXP-1 (BATCH SIZE 4)

Metric	VGG (%)	DenseNet (%)
Accuracy	69.50	71.73
Recall (macro)	69.50	70.56
Precision (macro)	68.87	69.36
F1-Score (macro)	64.12	69.72
ROC AUC	55.70	78.99

the following metrics: accuracy, precision, recall (sensitivity), F1-score, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). These metrics collectively provide a comprehensive assessment of model behavior on the imbalanced test set.

III. RESULT AND DISCUSSION

This section presents the results for the three configurations (Exp-1, Exp-2, and Exp-3) for each architecture applied to the local CXR dataset. Performance is assessed using a classification report, ROC curve, and training curve for each test. In the last part of this section, we discuss the findings and the limitations of this study.

A. Experiment 1 (batch size 4)

The first experiment (Exp-1) was conducted using a batch size of 4 with a learning rate of 5×10^{-5} for VGG19, and a learning rate of 3×10^{-4} with a ReduceLROnPlateau scheduler for DenseNet121, as detailed in Section II-B. For DenseNet121, training accuracy increased gradually from approximately 57% to 68%, while validation accuracy fluctuated heavily during the early epochs before stabilizing in the range of 58–65% (Fig. 6). The gap between training and validation accuracy remained relatively small throughout training, indicating that overfitting in this configuration was mild and the model retained reasonable generalization capability. The test results, presented in Table II, show that DenseNet121 achieved an accuracy of 71.73% and a ROC-AUC of 0.7899, substantially outperforming VGG19’s ROC-AUC of 0.5570 in this configuration. Meanwhile, the VGG’s results exhibited fluctuations in validation accuracy throughout the 30-epoch training period, indicative of erratic gradient updates characteristic of very small batch sizes. The validation accuracy demonstrated high volatility, oscillating between approximately 35% and 68%. In contrast to DenseNet121, the VGG’s training accuracy remained stagnant at around 50%. For this configuration, the DenseNet model is better suited for CXR classification than the VGG model.

TABLE III
EVALUATION REPORT FOR EXP-2 (BATCH SIZE 8)

Metric	VGG (%)	DenseNet (%)
Accuracy	59.75	73.68
Recall (macro)	59.75	74.93
Precision (macro)	68.31	72.63
F1-Score (macro)	60.44	72.65
ROC AUC	72.57	82.21

B. Experiment 2 (batch size 8)

The second experiment (Exp-2) was conducted using a batch size of 8, with the same training configurations as Exp-1 for each architecture. For DenseNet121, training accuracy increased steadily from approximately 62% to 78%, while validation accuracy plateaued around 63–69% after an initial sharp rise (Fig. 6). A clearer divergence pattern emerged in this configuration compared to Exp-1, with the train-validation gap widening progressively from around epoch 6 onward, indicating that moderate overfitting occurred. The test results in Table III show that DenseNet121 achieved an accuracy of 73.68% and a ROC-AUC of 0.8221, outperforming VGG19 across all metrics for this batch size. Meanwhile, the VGG’s validation accuracy fluctuated throughout the 30-epoch training period, oscillating between approximately 35% and 55%, while its training accuracy remained stagnant at around 50%. This suggests that the DenseNet model is more suitable for classifying CXRs than the VGG model, as it outperforms on all metrics.

C. Experiment 3 (batch size 16)

The third experiment (Exp-3) was conducted using a batch size of 16, with the same training configurations as the previous experiments for each architecture. For DenseNet121, training accuracy increased sharply and continuously, reaching approximately 91% by the final epoch, while validation accuracy remained largely stagnant in the range of 65–70% (Fig. 6). This configuration exhibited the widest and most persistent train-validation gap among the three batch sizes, indicating that the larger batch size, combined with the increased model capacity from pretrained initialization, allowed the model to fit the training data increasingly well without a corresponding improvement in generalization. Despite this overfitting behavior, the test results in Table IV show that DenseNet121 achieved its highest accuracy (75.07%) and ROC-AUC (0.8229) among the three configurations, suggesting that the best-performing checkpoint—selected based on validation accuracy—was captured before overfitting became severe. Meanwhile, the VGGs demonstrated comparatively smoother convergence behavior. While validation metric fluctuations remained, the generalization gap was more moderate than in the smaller-batch-size experiments, with validation accuracy oscillating between approximately 35% and 60% and training accuracy stagnant around 55%. With pretrained initialization, DenseNet121 outperforms VGG19 across all metrics at batch size 16.

TABLE IV
EVALUATION REPORT FOR EXP-3 (BATCH SIZE 16)

Metric	VGG (%)	DenseNet (%)
Accuracy	66.990	75.07
Recall (macro)	66.99	73.39
Precision (macro)	69.37	72.57
F1-Score (macro)	67.68	72.90
ROC AUC	73.50	82.29

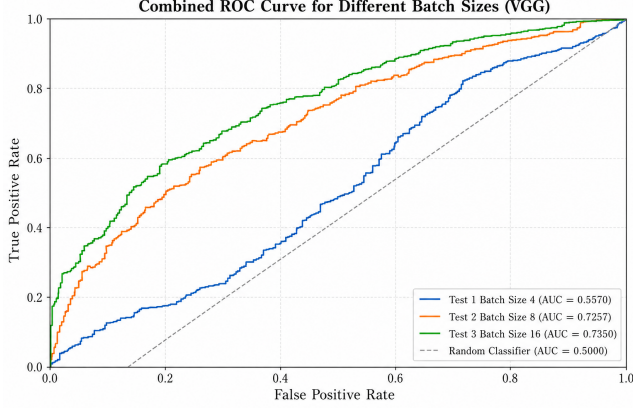


Fig. 4. Batch Size Effect on ROC-AUC (VGG).

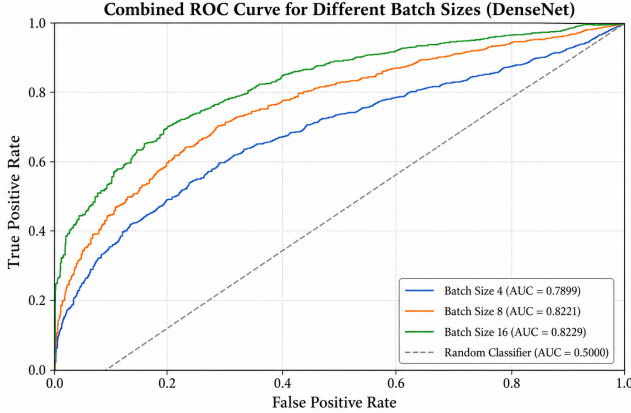


Fig. 5. Batch Size Effect on ROC-AUC for DenseNet121 (Pretrained).

D. Discussion

The results of this study reveal several important findings regarding the interplay between mini-batch size, training dynamics, and the clinical adequacy of the resulting TB detection model. Three principal axes of analysis are addressed: the clinical implications of the sensitivity-specificity trade-offs, the mechanistic explanation for the batch size effects, and the overfitting behavior observed during training.

1) *Clinical Implications of the Sensitivity-Specificity Trade-Off:* In medical diagnostics, the relative costs of false negatives and false positives are rarely equal. For a disease such as TB, which is both infectious and life-threatening if untreated, the clinical imperative to minimize false negatives and missed TB cases is paramount. A missed TB diagnosis delays treatment

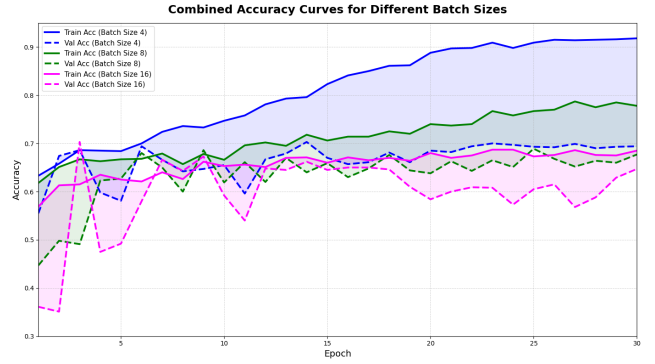


Fig. 6. Combined training and validation accuracy curves for DenseNet121 across the three batch-size configurations (4, 8, and 16), illustrating the progressive widening of the train-validation gap with increasing batch size.

initiation, exposes close contacts to ongoing transmission risk, and contributes to the emergence of drug-resistant strains [1]. Viewed through this lens, the performance of Exp-1 (batch size = 4) presents a paradox. While the VGG's accuracy of 69.50% appears acceptable, the class-level analysis reveals that the model is critically biased toward predicting the majority class. The non-TB recall of 22.27% implies that 192 of 247 non-TB patients were incorrectly classified as TB-positive, generating an unacceptably high false positive burden. From a health systems perspective, this would translate to substantial overload of confirmatory diagnostic procedures, unnecessary patient anxiety, and unjustified initiation of anti-TB therapy with its associated toxicity profile [3].

Overall, the comparative analysis between VGG19 and DenseNet reveals that both architectures exhibit substantial trade-offs between TB sensitivity and non-TB specificity, highlighting the clinical complexity of deploying AI-based TB screening systems. The VGG19 model demonstrated a more balanced classification behavior as the batch size increased, with batch size 16 emerging as the most clinically acceptable VGG19 configuration. This model achieved relatively equitable recall values for both the TB and non-TB classes, along with a ROC-AUC score of 73.50%, indicating reasonable discriminative capability, illustrated in Fig. 4. From a medical perspective, this balance is important because it reduces the risks of both excessive false positives, which may overload healthcare systems and expose healthy individuals to unnecessary follow-up procedures, and false negatives, which are considerably more dangerous because they can lead to missed active TB cases and continued disease transmission [13], [17].

In contrast to the original from-scratch configuration, the pretrained DenseNet121 exhibited consistent improvement in discriminative capability as batch size increased, as illustrated in Fig. 5. Batch size 16 achieved the highest ROC-AUC (0.8229), followed by batch size 8 (0.8221) and batch size 4 (0.7899). However, class-level analysis indicates that DenseNet121 retains a tendency toward TB-positive prediction bias across all batch sizes: at batch size 16, the non-TB recall

was 68.02% with a TB recall of 78.77%, while at batch size 4, the non-TB recall was 66.80% with a TB recall of 74.31%. Although less severe than in the original from-scratch configuration—where non-TB recall dropped as low as 19% at batch size 16—this persistent asymmetry indicates that, despite the substantial improvement in overall discriminative capability following pretrained initialization and the application of class-weighting strategies, a residual class imbalance bias remains. This suggests that further refinement of the classification threshold or additional class-balancing techniques may be warranted to achieve more equitable sensitivity-specificity trade-offs suitable for clinical screening contexts. These findings collectively suggest that, in TB screening applications, overall accuracy alone is insufficient for evaluating model performance. Instead, clinically meaningful metrics such as class-specific recall, false-positive burden, and false-negative risk must be prioritized, as they directly influence patient safety, diagnostic workload, treatment appropriateness, and public health control strategies.

2) *Mechanistic Analysis of Batch Size Effects:* The observed differences across the three experiments for VGG19 can be explained by the influence of mini-batch size on stochastic gradient descent dynamics. In Exp-1 (batch size = 4), the extremely small batch size produced highly noisy and unstable gradient updates, resulting in erratic traversal of the loss landscape and unstable validation performance. Under the imbalanced dataset condition, this noisy optimization process biased the model toward the majority TB class, leading to very high TB sensitivity but extremely poor non-TB specificity [17]. Consequently, the model achieved seemingly acceptable overall accuracy while failing to learn robust discriminative representations for both classes. Exp-2 (batch size = 8) provided slightly more stable gradient estimation, but the optimization process remained inconsistent, resulting in a different failure mode characterized by excessive false negatives and clinically concerning missed TB detections.

In contrast, Exp-3 (batch size = 16) demonstrated that larger batches yielded smoother, more representative gradient estimates for VGG19, enabling more stable convergence and improved exploration of discriminative features across both classes. This resulted in the most balanced recall performance and the highest ROC-AUC value among the VGG19 configurations, indicating superior generalization capability relative to the smaller-batch-size VGG19 experiments. From a clinical perspective, these findings suggest that appropriate batch size selection is critical not only for improving model accuracy but also for reducing diagnostically dangerous prediction bias that may either overwhelm healthcare systems through false positives or compromise patient safety through false negatives. These observations collectively underscore that hyperparameter selection, specifically batch size, is not merely a computational consideration but a critical determinant of the clinical safety profile of the resulting model. In the context of imbalanced medical datasets, the interaction between batch size and class distribution can fundamentally alter the diagnostic bias of the deployed system.

3) *Overfitting Analysis:* Fig. 6 presents the combined training and validation accuracy curves for DenseNet121 across all three batch-size configurations, enabling direct visual comparison of the train-validation gap. A consistent pattern emerged in which the divergence between training and validation accuracy increased with batch size. At batch size 4, training and validation curves remained closely aligned throughout training, indicating mild overfitting. At batch size 8, a moderate gap emerged after approximately epoch 6, with training accuracy continuing to rise while validation accuracy plateaued. At batch size 16, this divergence became the most pronounced, with training accuracy approaching 91% while validation accuracy remained largely confined to the 65–70% range, indicating severe overfitting.

This trend can be attributed to the interaction between batch size, gradient estimate stability, and the use of ImageNet-pretrained weights. Larger batch sizes produce smoother and more consistent gradient updates, which—combined with the already rich feature representations from pretrained initialization—allow the model to rapidly fit the training distribution, including dataset-specific noise and artifacts. In contrast, smaller batch sizes introduce more stochastic noise into the optimization process, which can act as an implicit regularizer, slowing the rate at which the model memorizes training-specific patterns.

Importantly, despite the presence of more pronounced overfitting at batch sizes 8 and 16, both configurations achieved higher test accuracy and ROC-AUC than batch size 4. This is because the best-performing model checkpoint was selected based on validation accuracy at each epoch (model checkpointing), rather than using the final-epoch weights. As a result, the reported test performance reflects the model state before overfitting became dominant. This finding highlights the importance of checkpoint selection and validation-based early stopping as practical mitigation strategies, and motivates their inclusion as future work (Section IV).

The application of data augmentation (Section II-B-1), class-weighted loss, and a weighted random sampler likely contributed to mitigating—though not eliminating—overfitting, particularly at smaller batch sizes. The residual overfitting observed at larger batch sizes suggests that additional regularization strategies, such as stronger dropout, weight decay tuning, or explicit early stopping based on a patience criterion, would likely yield further improvements in generalization.

4) *Low Model Performance Discussion:* Although the revised DenseNet121 experiments demonstrate notable improvements over the original from-scratch configurations—with ROC-AUC increasing from a range of 0.557–0.746 to 0.790–0.823—the resulting accuracy (71.7–75.1%) and ROC-AUC values remain below thresholds typically considered acceptable for standalone clinical deployment. Several factors likely contribute to this performance ceiling. First, the local dataset, comprising 4,784 images collected from a single pulmonary hospital, is relatively limited in scale and may not capture the full variability of TB and non-TB presentations encoun-

tered across different populations, imaging equipment, and acquisition protocols. Second, chest X-ray images sourced from real clinical settings often exhibit substantial variability in image quality, contrast, positioning, and the presence of confounding pathologies, which can obscure TB-specific radiological features and increase classification difficulty. Third, TB diagnosis from CXR alone is inherently challenging even for experienced radiologists due to overlapping radiological presentations with other pulmonary conditions, contributing to label noise and irreducible classification uncertainty. Fourth, this study represents a preliminary investigation focused primarily on architectural comparison and batch-size sensitivity, rather than an exhaustive optimization of the training pipeline; consequently, the reported performance should be interpreted as a baseline rather than an upper bound. Finally, the development of robust AI-based diagnostic tools using local Indonesian datasets remains an emerging area, where challenges related to dataset size, annotation consistency, and population-specific disease presentation patterns are still being characterized. These findings underscore that, while the comparative trends between architectures are informative, further work involving larger and more diverse datasets, additional regularization, and rigorous external validation is necessary before such models could be considered for clinical screening applications.

5) *Limitation Statement*: It is acknowledged that, while both VGG19 and DenseNet121 in this study were initialized with ImageNet-pretrained weights, ensuring a consistent initialization strategy, the two architectures were trained under different optimization configurations. VGG19 was trained using a fixed learning rate of 5×10^{-5} without a learning rate scheduler, data augmentation, or class-weighting strategies, whereas DenseNet121 was trained using a learning rate of 3×10^{-4} with a ReduceLROnPlateau scheduler, data augmentation, class-weighted loss, and a weighted random sampler. Consequently, the performance differences reported between the two architectures may reflect, in part, differences in training configuration rather than purely architectural characteristics. Future studies should evaluate both architectures under identical training protocols—including learning rate, scheduler, augmentation, and class-balancing strategies—to enable a more controlled and architecture-isolated comparison.

IV. CONCLUSION

This study has demonstrated that the performance comparison between the VGG19 and DenseNet121 CNN architectures for TB detection from CXR images is sensitive to both batch size and training configuration. With both architectures initialized using ImageNet-pretrained weights, DenseNet121 consistently outperformed VGG19 across all three batch-size configurations in terms of accuracy and ROC-AUC, achieving ROC-AUC values between 0.790 and 0.823 compared to 0.557–0.735 for VGG19. Analysis of training and validation curves revealed that overfitting in DenseNet121 became more pronounced as batch size increased, although validation-based checkpoint selection allowed the best-performing models to

be captured prior to severe overfitting. Despite these improvements, overall performance remains below the threshold required for standalone clinical deployment, which can be attributed to the limited size and variability of the local dataset, the inherent difficulty of TB classification from CXR, and the preliminary nature of this study.

Future Work

Future research directions include: (1) systematic learning rate optimization, including comparative evaluation of different initial learning rates and optimizers; (2) more extensive use of learning rate schedulers, such as cosine annealing, in combination with explicit early stopping based on validation loss patience; (3) fine-grained tuning of weight decay and dropout rates to further mitigate overfitting, particularly at larger batch sizes; (4) evaluation of both architectures under identical training protocols to isolate architectural effects from optimization effects; (5) exploration of metaheuristic optimization approaches, such as Particle Swarm Optimization (PSO) and Grey Wolf Optimizer (GWO), for automated hyperparameter tuning; and (6) validation on larger and more diverse local and multi-center datasets to improve generalizability and clinical applicability. Comparative studies involving alternative architectures, such as ResNet, EfficientNet, and Vision Transformers, on the same dataset are also warranted to establish the relative advantages of these architectures in this clinical context.

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